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INFORMATION TECHNOLOGY

Software Requirements Specification(SRS)

Report On

"India Insider"

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1. Introduction

1.1 System Purpose

The purpose of the **India Insider** system is to create a digital tourism platform that provides comprehensive and interactive information about India's tourist destinations, cultural heritage, events, and attractions. The system aims to promote tourism through digital media by offering a one-stop solution for travelers to explore destinations, plan itineraries, and access travel-related details conveniently from any device.

This platform bridges the gap between travelers and Indian tourism resources, simplifying travel research, planning, and discovery through an intuitive web interface.

1.2 System Scope

The **India Insider** system is a **web-based application** that serves both domestic and international travelers.

The main features include:

- A detailed directory of **tourist places, hotels, restaurants, and cultural events**.
- **Interactive maps** using Google Maps API to locate destinations.
- **Category-based search and filtering** (Adventure, Heritage, Wildlife, Religious, Beaches, etc.).
- **Dynamic content management**, allowing updates of new destinations.
- A user-friendly interface accessible on desktops and mobile devices.
- Future integration of **AI-based trip suggestions** and **chatbot support**.

The scope covers:

- **Frontend Design:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** PHP / Node.js
- **Database:** MySQL / MongoDB
- **Deployment:** Web server (Firebase / GitHub Pages / Netlify)

1.3 Reports & Analytics

The system will generate:

1. **User Analytics Reports:**
   Tracking website visitors, page views, and user activity.

2. **Destination Popularity Reports:**
   Insights into most-searched or most-visited places.

3. **Feedback Reports:**
   Analysis of user ratings, comments, and satisfaction levels.

4. **Search Statistics:**
   Trending destinations and seasonal interests.

These analytics help in improving the website's content and user experience.

2. System Overview

2.1 System Context

The system functions as a **centralized online platform** for showcasing Indian tourism.
Users interact with the website via browsers. The backend server processes requests, retrieves destination data from the database, and displays it to the user. External APIs (like Google Maps and weather services) enhance interactivity.

Context Diagram:

```
+-------------------------+
|       User / Tourist    |
+-----------+-------------+
            |
            v
+-------------------------+
|  India Insider Website  |
| (Frontend + Backend)    |
+-----------+-------------+
            |
   ------------------------
   |        |             |
   v        v             v
+--------+ +--------+ +-------------+
|Database| |Google  | |Weather API  |
| (MySQL)| | Maps   | | Integration |
+--------+ +--------+ +-------------+
```

2.2 System Functions

The core functionalities of the India Insider system include:

1. Displaying a categorized list of tourist destinations.
2. Showing maps and directions for each place.
3. Allowing users to filter destinations by region or theme.
4. Displaying hotel, restaurant, and transport options nearby.
5. Managing and storing destination data in the backend database.

6. Generating analytics on user activity and destination popularity.

2.3 User Characteristics

- **Admin Users:**
  Manage destination content, upload new locations, update images, and monitor reports.
  Require basic technical knowledge of website operations.

- **Tourists / General Users:**
  Browse, search, and explore content.
  Need no technical expertise; simple navigation suffices.

3. Definitions

Term	Definition
Frontend	User interface part of the website.
Backend	Server-side logic that handles requests and data management.
API	Application Programming Interface used for connecting to external services (e.g., Google Maps).
MySQL	A relational database used to store tourism data.
Responsive Design	Website layout that adjusts automatically to different screen sizes.
Tourism Analytics	Reports on user activity and destination engagement.

4. Requirements

4.1 External Interfaces

- **Google Maps API** for displaying maps and routes.
- **Weather API** to show live weather at destinations.
- **Payment Gateway (future)** for booking and ticket integration.

4.2 Hardware Interface

Component	Description
Server	Cloud-hosted web server (e.g., Firebase, AWS)
Client Device	Desktop, Laptop, Smartphone
Internet	Required for accessing the website and APIs

4.3 System Interface

- Communication between frontend and backend via **HTTP / HTTPS**.
- Backend retrieves data from the **MySQL** database.
- External APIs integrated using **RESTful endpoints**.

4.4 User Interface

- **Home Page:** Introduction and featured destinations.
- **Destination Page:** Detailed information, maps, and photos.
- **Search/Filter Page:** Sort destinations by category, region, or theme.
- **Admin Dashboard:** For managing and updating content.

5. Performance Requirements

1. The website must load within **3–4 seconds** on standard connections.
2. It should support **minimum 100 concurrent users**.
3. Database queries should execute in less than **2 seconds**.
4. System uptime should be **99%**.
5. The interface should be responsive on both desktop and mobile browsers.

6. Design Constraints

- Must use **open-source technologies** (HTML, CSS, JS, PHP).
- Must ensure **cross-browser compatibility** (Chrome, Edge, Firefox).
- Limited **storage space** on hosting platform.
- API usage restricted by **free-tier limits** (Google Maps).
- Security compliance with **HTTPS and input validation**.

7. Functions

- Manage destinations (CRUD operations for admin).
- Search and display places by name, state, or category.
- Display location using Google Maps integration.
- Generate travel suggestions and nearby attractions.
- Store and retrieve user activity for analytics.

8. Functional Requirements

1. System should display a list of destinations with relevant details.
2. Users should be able to view maps, photos, and key information.
3. The admin should add, edit, and delete destinations.
4. The system should store all records in the database.
5. System should fetch and display weather data for destinations.
6. Website should allow feedback and ratings (optional future feature).

9. Non-Functional Requirements

Type	Requirement
Performance	Fast loading, low latency
Reliability	24x7 uptime
Usability	Simple interface, easy navigation
Scalability	Should handle more destinations and users in future
Security	Use of HTTPS, SQL injection prevention
Maintainability	Modular code structure for easy updates
Portability	Accessible from any web browser or device

10. Use Case Modelling

10.1 System Flow Diagram

```
        +------------------------+
        |     User / Visitor     |
        +-----------+------------+
                    |
                    v
        +------------------------+
        | India Insider Website  |
        |  (Frontend Interface)  |
        +-----------+------------+
                    |
          ------------------------
          |         |            |
          v         v            v
   +-----------+ +--------+ +----------+
   | Destination| | Map API| | Database |
   |   Module   | |Weather | | MySQL DB |
   +-----------+ +--------+ +----------+
                    |
                    v
        +------------------------+
        |  Output to User (Web)  |
        +------------------------+
```

10.2 Data Flow Diagram (Level 0)

```
        +----------------+
        |     User       |
        +-------+--------+
                |
                v
    +----------------------+
    |   India Insider Web  |
    |   Application (UI)   |
    +-------+--------------+
            |
    -------------------------
    |           |            |
    v           v            v
+---------+ +-----------+ +------------+
|Database | | Maps API  | | Weather API|
+---------+ +-----------+ +------------+
```